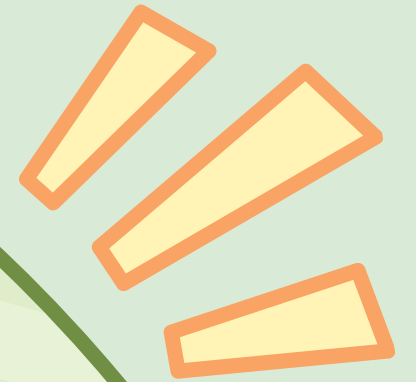




Soil Layers – Make a Core Sample



What We Will Learn Today

Today we will discover exciting things about soil!

Name the 3 layers of soil

Make our own core sample

Discover what lives in soil

Get ready to become soil scientists!





What is Under Our Feet?

Have you ever wondered what is below the ground where we play? Let's find out together!



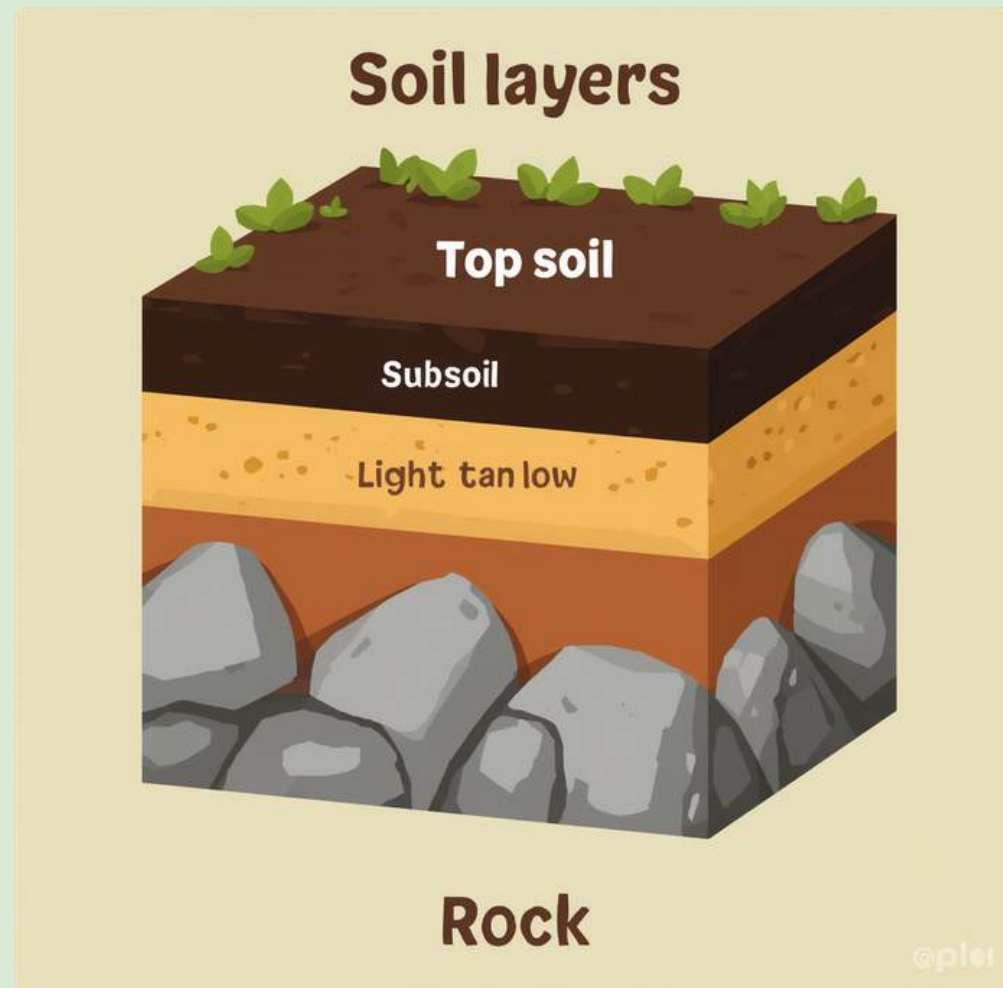


Soil is Like a Layered Barfi!

Just like barfi has different colored layers, the ground also has layers! Each layer is different and special. Look at the picture — can you see how the sweet and the soil both have layers stacked on top of each other?



The 3 Layers of Soil



- TOP LAYER (Topsoil) — Dark and soft, this is where plants grow and tiny creatures live! It feels crumbly like chocolate cake.
- MIDDLE LAYER (Subsoil) — Light brown or yellow colored, sticky like clay. Big tree roots reach down here to drink water!
- ROCK LAYER (Parent Rock) — Hard stones and pebbles at the very bottom. This is the strong foundation that holds everything up!





Top Layer – Topsoil



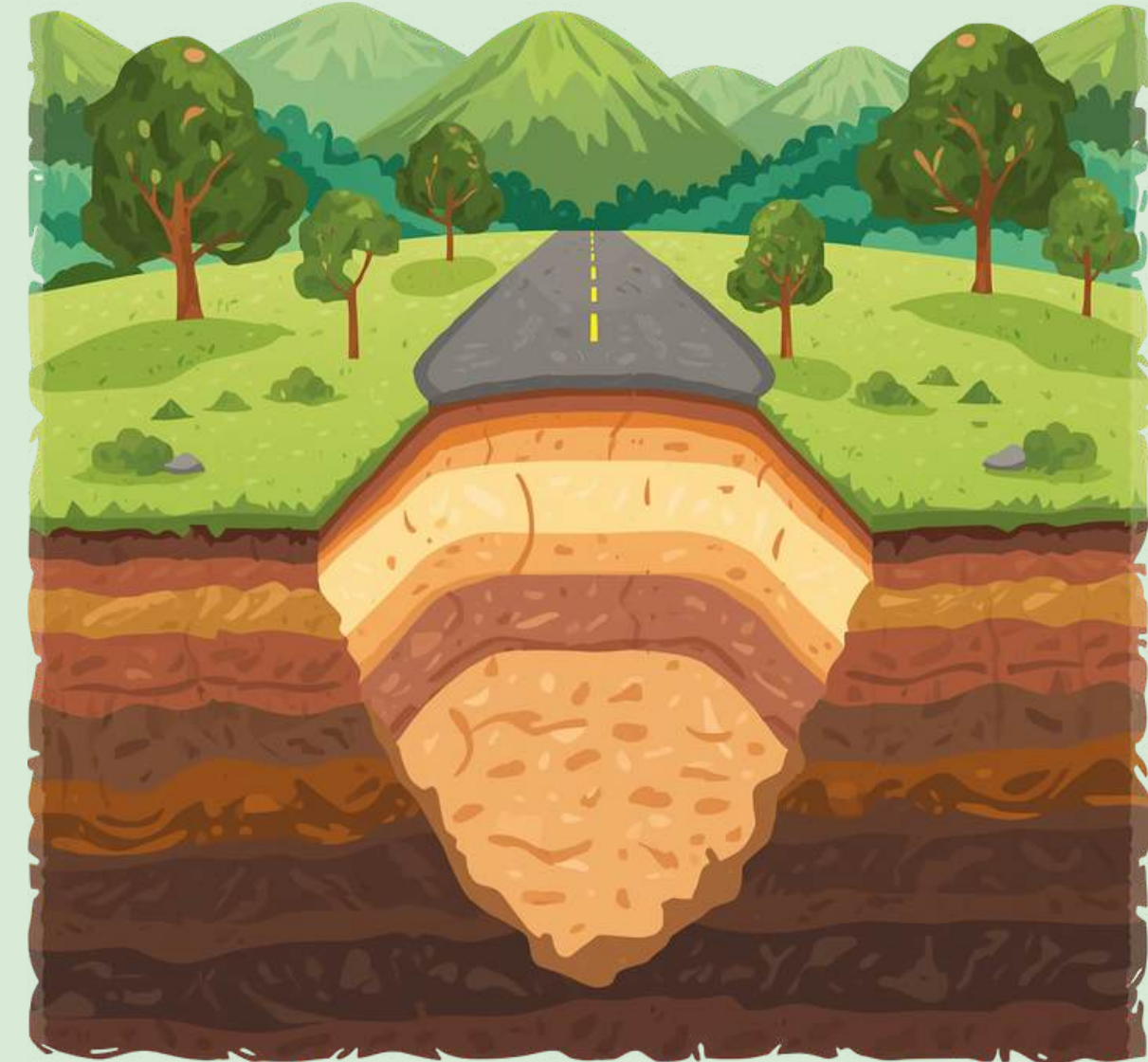
- Dark brown or black color — like the rich soil in your school garden!
- Soft and crumbly — easy to dig with your fingers
- Where plants grow and worms live! The roots of tulsi, mango trees, and vegetables all start here. Earthworms make their homes in topsoil too!





Middle Layer — Subsoil

- Lighter color — yellow or light brown
- More clay, sticky and dense
- Big tree roots reach here!
- Indian example: Lighter soil visible in road cutting in Kerala or Karnataka hills





Rock Layer — Parent Rock

- Grey or tan color — looks like sand and stones mixed together
- Hard pebbles and stones that feel rough
- The foundation of everything above — like the bottom of a building!

Indian Example: You can see rocky pieces at construction sites or on the famous Deccan Plateau where big rocks sit under the soil!





What is a Core Sample?

A core sample is like pushing a straw into a layered sandwich! Scientists use it to see all the layers without digging a big hole. Instead of digging up the whole ground, we can take a small sample and see everything inside!





Let's Make Our Core Sample!



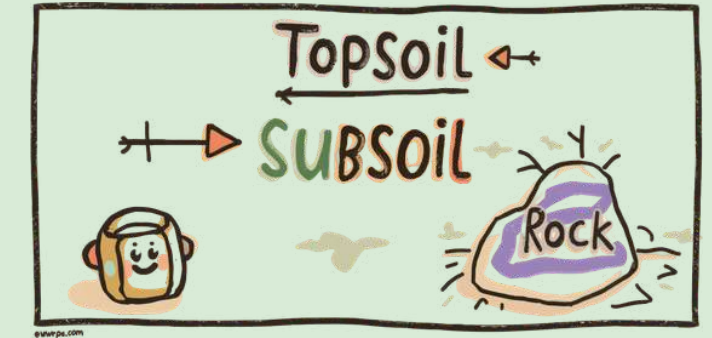
Step 1: Add gravel and pebbles at the **BOTTOM** of your bottle. This is the Rock Layer — hard and strong!

Step 2: Add light sand or clay in the **MIDDLE**. This is the Subsoil — lighter and sticky!





Let's Make Our Core Sample! (Part 2)



Step 3: Add Dark Soil on TOP
This is the TOPSOIL layer!
It's dark brown and soft.
Put it right on top of your middle layer. This is where plants grow!

Step 4: Add a Tiny Decoration
Place a small leaf or a toy worm on top! This shows what lives in topsoil.
Earthworms love it here!

Step 5: Label Each Layer!
Use stickers or paper to label: TOPSOIL at top, SUBSOIL in middle, ROCK at bottom. Your core sample is complete!



What Lives in Topsoil?



- Earthworms — Farmer's friends! They dig tunnels and make soil healthy for plants to grow.
- Ants — Tiny tunnel builders! They carry food underground and help mix the soil layers.
- Plant roots — Mango, neem, and tulsi roots drink water and hold tight to the topsoil!





Let's Remember Together!

What's on TOP?

→ TOPSOIL! Dark and soft!

What's in the MIDDLE?

→ SUBSOIL! Clay and light!

What's at the BOTTOM?

→ ROCK! Hard and tight!

What do we call our science tube?

→ A CORE SAMPLE! Wah!





Take This Home!

Tonight, ask someone at home:
"When we plant tulsi or
tomatoes, which layer do the
seeds go in?"
The answer is... TOPSOIL!
Can you explain why?





**Thank You for Learning
About Soil Layers!**

